



Software Testing and Quality Assurance

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Module-3

Types of Testing

- Installation testing,
- Usability testing,
- Regression testing,
- Performance testing,
- Load testing,
- Stress testing,
- Security testing,
- Case study on types of testing.

Installation testing

Installation Testing is a type of software testing that verifies whether a software product is correctly installed, configured, and uninstalled on a target environment. It ensures that the installation process proceeds smoothly, the software is set up properly, and users can run the program without encountering issues due to improper installation.

Definition:

Installation Testing focuses on the processes involved in setting up and deploying software, ensuring that installation procedures work as intended and that the software integrates well with the operating environment.



Key Objectives:

Correct Setup: Confirm that software installation (and subsequent uninstallation) completes without errors.

Configuration Validation: Verify that all components—files, directories, registry settings (on Windows), environment variables—are correctly configured.

Environment Compatibility: Ensure that the software installs successfully across different environments, operating systems, and hardware configurations.

User Experience: Validate that the installation process is user-friendly and adheres to any provided guidelines or instructions.

Installation Testing Process:

1. Pre-Installation Checks:

- Verify system prerequisites (OS version, available disk space, necessary libraries, etc.).
- Ensure compatibility with different hardware or virtual environments.

2. Installation Execution:

- Run the installation process using various methods (e.g., installers, setup wizards, scripts).
- Monitor for any errors, warnings, or misconfigurations during installation.



3.Post-Installation Verification:

- Confirm that all files, directories, and configuration settings have been correctly deployed.
- Check that the software launches properly and performs basic functionality tests.
- Validate the uninstall process to ensure that no residual files or settings remain after removal

4.Error Handling & Recovery:

- Test how the installer handles installation failures (e.g., corrupt files, interrupted installations).
- Verify that appropriate error messages are presented and that the software can recover or roll back gracefully.



Benefits:

- **Reliability Assurance:** Ensures the installation process is robust, reducing the chances of issues in production.
- **User Satisfaction:** Delivers a smooth setup experience, enhancing user trust and satisfaction.
- **Environment Consistency:** Confirms that the software performs consistently across various user systems and configurations.
- **Quality Control:** Helps identify issues early in the deployment cycle before reaching end-users.



Challenges:

- **Diverse Environments:** Testing across different operating systems, configurations, and hardware settings can be complex.
- **Automation Limitations:** Some parts of the installation process may be difficult to automate, requiring manual testing.
- **Dependency Management:** Verifying that all necessary dependencies and prerequisites are correctly handled during installation can be challenging.

Integration with Other Testing Levels:

Installation Testing is usually part of the broader release or system testing process. While unit testing, integration testing, and system testing focus on the functionality and interaction of software components, installation testing ensures that the software, when deployed, provides a seamless experience from setup to execution.

How to Perform Installation Testing

Requirements

Test Environment

Design Test Cases



Execute Test Cases

Document

Repeat



Usability testing

Usability testing is a method used to evaluate the **user experience** and **navigation** of websites, apps, and digital products.

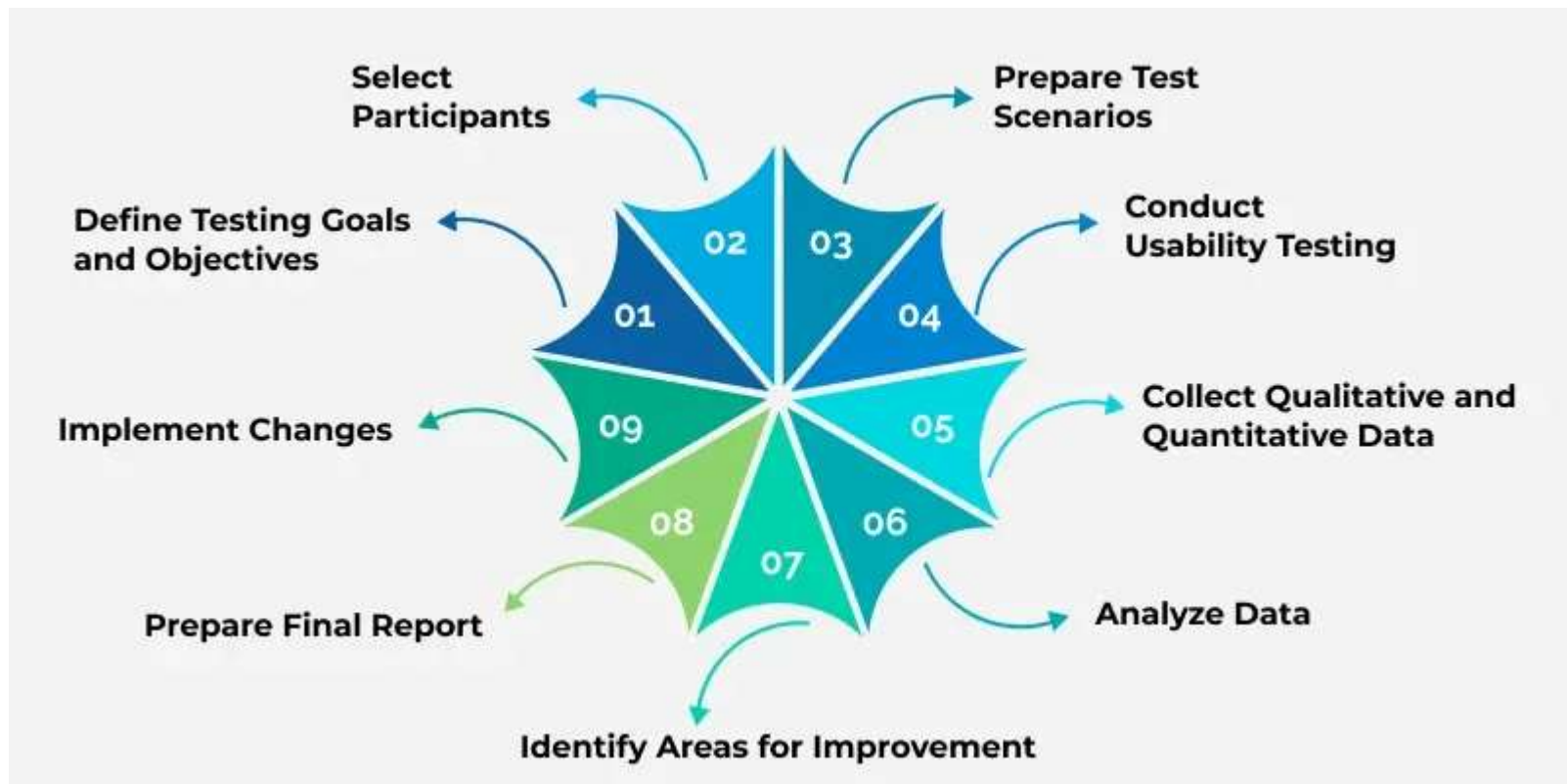
In this guide, we'll explore the basics of usability testing, its significance in **software development**, and how it enhances **user engagement**. Whether it's through in-person sessions or remote testing, we'll delve into how real users' interactions provide invaluable insights for **product improvement**.



Usability testing

Usability Testing in software testing is a type of testing, that is done from an end user's perspective to determine if the system is easily usable. Usability testing is generally the practice of testing how easy a design is to use on a group of representative users. Several tests are performed on a product before deploying it. You need to collect **qualitative and quantitative data** and satisfy customers' needs with the product. A proper final report is made mentioning the changes required in the product (software).

Usability testing



Regression testing

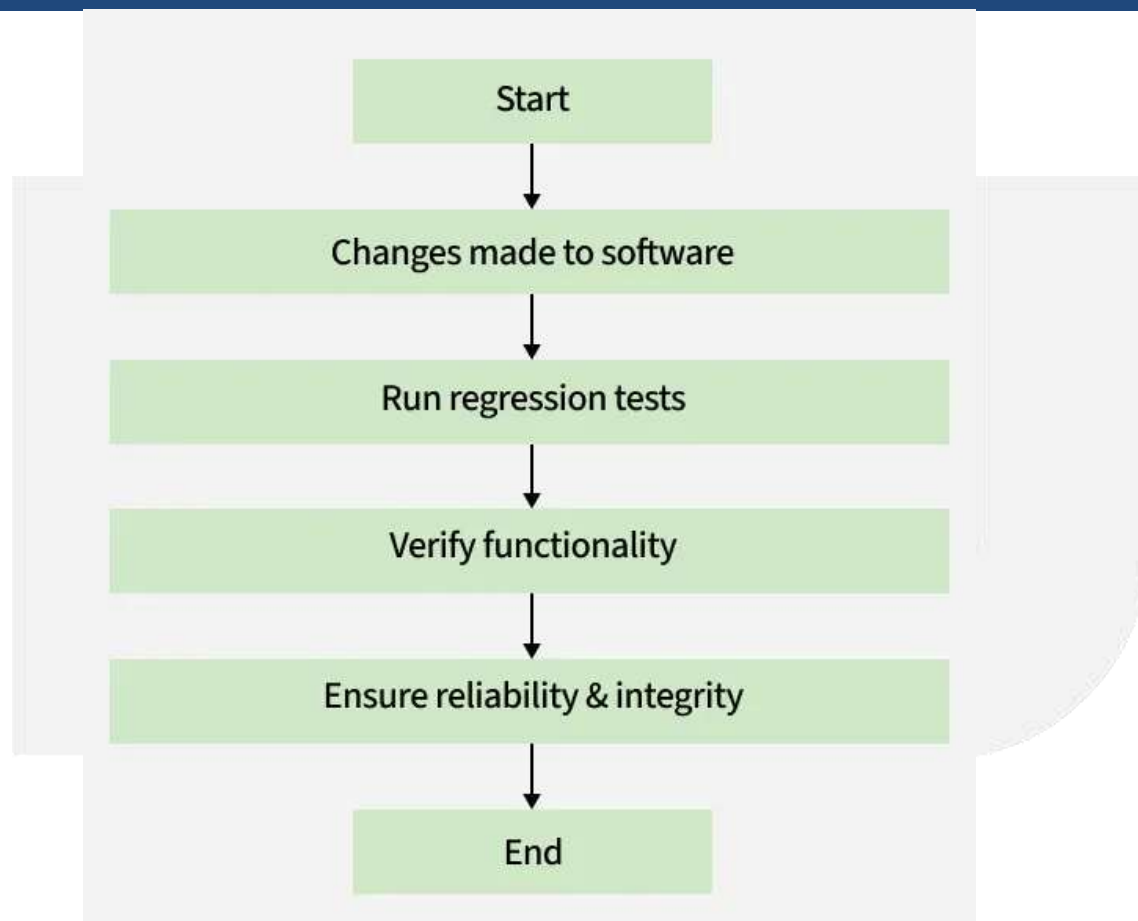
Regression testing is the process of re-running test cases to ensure recent code changes have not broken existing functionality. For example, a calculator application supports addition, subtraction, multiplication, and division.

Regression testing

Regression testing is necessary in several scenarios to maintain software quality:

- When new functionality is added to the system and the code has been modified to absorb and integrate that functionality with the existing code.
- When some defect has been identified in the software and the code is debugged to fix it.
- When the code is modified to optimize its working.

Regression testing



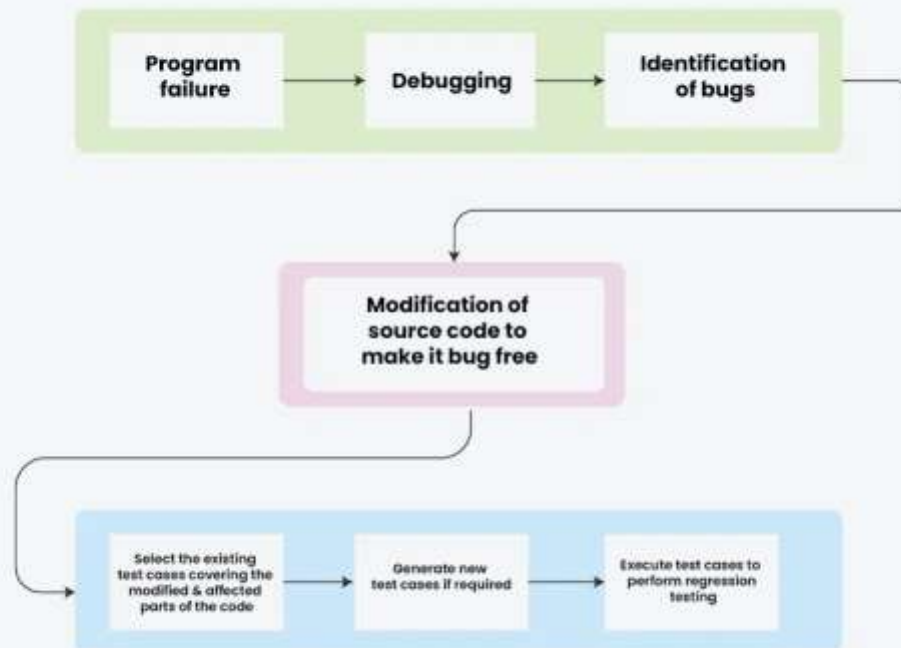
Regression testing

Process of Regression testing

Firstly, whenever we make some changes to the source code for any reason like adding new functionality, optimization, etc. then our program when executed fails in the previously designed test suite for obvious reasons. After the failure, the source code is debugged to identify the bugs in the program. After identification of the bugs in the source code, appropriate modifications are made.

Regression testing

Process Regression testing





Regression testing

Selection of Test cases for Regression Testing



Prioritization

On the basis of technical,
requirements



Priority Code 1

Essential Test Case

Priority Code 2

Important Test Case

Priority Code 3

Execute, if resources permits

Priority Code 4

Not important Test Case

Priority Code 5

Redundant Test Case

On the basis of customer
requirements



Priority Code 1

Important for the customer

Priority Code 2

Required to increase the
customer satisfaction

Priority Code 3

Helps to increase the market
share of the product

Top Regression Testing Tools

1. Selenium

Open Source: Selenium is an open-source tool, making it freely available and accessible for developers and testers.

Browser Compatibility: Supports multiple browsers, including Chrome, Firefox, Safari, and Edge, ensuring tests can be run across different environments.

Programming Language Support: Allows writing tests in various programming languages such as Java, Python, C#, Ruby, and JavaScript, providing flexibility for testers.

Cross-Platform: Capable of running on different operating systems, including Windows, macOS, and Linux, which enhances the tool's portability.

Web Application Testing: Primarily designed for automating web applications, making it ideal for regression testing of web-based systems.

Extensive Community Support: Boasts a large and active community, offering a wealth of resources, plugins, and extensions to aid in test automation.

Integration Capabilities: Integrates well with other tools such as Jenkins for continuous integration and continuous deployment (CI/CD), facilitating automated regression testing in development pipelines.



Top Regression Testing Tools

2. Ranorex Studio

- **Comprehensive Testing Solution:** Ranorex Studio provides a complete testing solution that supports end-to-end regression testing, including both functional and non-functional tests.
- **User-Friendly Interface:** The tool offers an intuitive and user-friendly interface, making it accessible for both beginners and experienced testers.
- **Cross-Platform Testing:** It supports cross-platform testing, enabling tests to be executed on desktop, web, and mobile applications, ensuring broad test coverage.
- **Codeless Test Automation:** Ranorex Studio offers codeless automation through its capture-and-replay functionality, allowing testers to create automated tests without extensive programming knowledge.
- **Robust Reporting:** The tool provides detailed and customizable test reports, helping teams to easily identify issues and track the quality of the software over time.
- **Integration Capabilities:** It integrates seamlessly with popular development and CI/CD tools such as Jenkins, Azure DevOps, and Git, promoting a streamlined workflow and continuous testing.
- **Data-Driven Testing:** Ranorex Studio supports data-driven testing, allowing testers to run the same set of tests with different data inputs, enhancing test coverage and reliability.



Top Regression Testing Tools

3. testRigor

AI-Powered Test Automation: testRigor utilizes artificial intelligence to automate the creation and maintenance of regression tests, reducing the effort and time required for manual testing.

Natural Language Processing (NLP): The tool allows users to write test cases in plain English, making it accessible to both technical and non-technical team members. This feature simplifies test creation and improves collaboration.

Codeless Test Creation: Users can create and execute test cases without writing any code, which speeds up the testing process and reduces the dependency on developers.

Cross-Browser and Cross-Platform Testing: testRigor supports testing across multiple browsers and platforms, ensuring that the software works consistently in different environments.

Self-Healing Tests: The tool automatically updates test scripts to adapt to minor changes in the application's UI, minimizing the maintenance overhead commonly associated with regression testing.



Top Regression Testing Tools

4. Sahi Pro

Cross-browser Testing: Supports multiple browsers, ensuring consistent performance across different environments.

Ease of Use: User-friendly interface with scriptless record and playback functionality, making it accessible for testers with varying levels of expertise.

Robust Reporting: Detailed reports and logs that help in tracking and analyzing test results efficiently.

Integrated Suite: Offers integration with various Continuous Integration (CI) tools like Jenkins, enhancing the automation workflow.

Scalability: Capable of handling large-scale test automation projects with ease.

Script Flexibility: Supports both scriptless testing and advanced scripting using JavaScript, providing flexibility for complex test scenarios.

Top Regression Testing Tools

5. Testlio

Global Network of Testers: Access to a diverse pool of professional testers from around the world. Ensures comprehensive test coverage across different devices, operating systems, and locations.

On-Demand Testing: Flexible scheduling allows for testing when needed, fitting into the development cycle seamlessly. Enables quick turnaround times for regression testing after code changes.

Integrated Platform: Combines test management, test execution, and reporting in a single platform. Simplifies the regression testing process and provides a centralized view of test results.

Comprehensive Reporting: Detailed reports with actionable insights. Helps identify and prioritize issues, making it easier to address regressions promptly.



Advantages of Regression Testing

- Regression testing offers benefits that enhance software quality and development efficiency:
- Automated unit testing.
- Comprehensive test coverage.
- System integration.
- Faster test execution completion.
- Improved developer productivity.
- Parallel testing.
- Reduced costs.
- Regression testing improves product quality.
- Reusability.
- Scalability.
- Time efficiency.

Disadvantages of Regression Testing

Disadvantages of Regression Testing

- Regression testing has some challenges which are as follows:
- It can be time and resource-consuming if automated tools are not used.
- It is required even after very small changes in the code.
- Time and Resource Constraints.
- Among the significant risks associated with regression testing are the time and resources required to perform it.
- Incomplete or Insufficient Test Coverage.
- False Positives and False Negatives.
- Test Data Management Challenges.

Performance testing

Performance testing is a type of software testing that focuses on evaluating the performance and scalability of a system or application. Performance testing aims to identify bottlenecks, measure system performance under various loads and conditions, and ensure that the system can handle the expected number of users or transactions.

Performance testing

Types of performance testing

Performance Testing is the process of analyzing the quality and capability of a product. It is a testing method to determine the system's performance in terms of speed, reliability, and stability under varying workloads.

Performance testing



Types of Performance testing



Performance testing

1. Load testing

Load testing simulates a real-world load on the system to see how it performs under stress. It helps identify bottlenecks and determine the maximum number of users or transactions the system can handle. It checks the product's ability to perform under anticipated user loads. The objective is to identify performance congestion before the software product is launched in the market.



Performance testing

2. Stress testing

Stress testing is a type of load testing that tests the system's ability to handle a high load above normal usage levels. It helps identify the breaking point of the system and any potential issues that may occur under heavy load conditions. It involves testing a product under extreme workloads to see whether it handles high traffic or not. The objective is to identify the breaking point of a software product.

Performance testing

3.Security Testing is a type of Software Testing that uncovers vulnerabilities in the system and determines that the data and resources of the system are protected from possible intruders.

It ensures that the software system and application are free from any threats or risks that can cause a loss.

Security testing of any system is focused on finding all possible loopholes and weaknesses of the system that might result in the loss of information of the organization.

Performance testing

Types of Security Testing

1 Vulnerability Scanning

2 Security Scanning

3 Penetration Testing

4 Risk Assessment

5 Security Auditing

6 Ethical Hacking

7 Posture Assessment

8 Application Security Testing

9 Network Security Testing

10 Social Engineering Testing



Types of Security Testing Tools

SAST

Static Application Security Testing

DAST

Dynamic Application Security Testing

IAST

Interactive Application Security Testing

SCA

Software Composition Analysis

MAST

Mobile Application Security Testing

RASP

Runtime Application Self-Protection

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